📘 **Revision Booklet: Science Exam 1 (Lessons 1–6)**
Name: _____ **Class:** _____

✅ **Exam Date: Monday**

📌 **Topics Covered:**

1. **Life Processes & Organs** (Lesson 1)
2. **Animal & Plant Tissues** (Lesson 2)
3. **Cells & Organ Systems** (Lesson 3)
4. **Mixtures & Solutions** (Lesson 4)
5. **Evaporation & Distillation** (Lesson 5)
6. **Chromatography** (Lesson 6)

🧬 **Lesson 1: Life Processes & Organs**

Key Concepts to Know:

- MRS GREN: The 8 life processes of living things
- The difference between living and non-living
- Organs in animals and plants
- Functions of the heart, lungs, liver, brain, roots, and leaves

Questions to Check Understanding:

1. What do the letters in MRS GREN stand for?
2. Which organ pumps blood around the body?
3. What is the function of plant leaves?
4. Name one organ in the human body and its function.
5. Do plants have organs? Explain.

Revision Task:
📄 Create a labeled diagram of a human and a plant, showing at least 3 organs in each and writing their functions.

🔬 **Lesson 2: Animal & Plant Tissues**

Key Concepts to Know:

- What tissues are: groups of similar cells working together
- Types of animal tissues: muscle, epithelial, connective, nervous

- Types of plant tissues: xylem, phloem, root hair
- How organs are made from different tissues

Questions to Check Understanding:

1. What is a tissue?
2. What type of tissue carries messages in the body?
3. Which tissue transports water in plants?
4. Name one connective tissue.
5. What tissues are found in the heart?

Revision Task:
🧠 Match organs to their correct tissue types using examples from slides. Practice drawing and labeling these tissues.

🧫 Lesson 3: Cells & Organ Systems

Key Concepts to Know:

- Differences between plant and animal cells
- Parts of cells: nucleus, cell membrane, cytoplasm, mitochondria, chloroplast, vacuole, cell wall
- How cells → tissues → organs → systems
- Organ systems and their functions (e.g. digestive, circulatory)

Questions to Check Understanding:

1. What structures are found only in plant cells?
2. What does the mitochondria do?
3. Put these in order: organ, tissue, system, cell.
4. What organs are in the respiratory system?
5. What does the circulatory system do?

Revision Task:
📋 Fill in a chart comparing parts of plant and animal cells (✓ or ✗ for each part).

🥤 Lesson 4: Mixtures & Solutions

Key Concepts to Know:

- What a mixture is and how it differs from a compound
- Solutions = solute + solvent

- Examples of solutions
- How to separate mixtures physically

Questions to Check Understanding:

1. Define solute and solvent.
2. Give an example of a solution.
3. What kind of mixture is saltwater?
4. How can you separate sand and water?
5. What method is used to recover the solute?

Revision Task:
📄 Make a table of examples: solute, solvent, and resulting solution (e.g. lemonade powder + water = lemonade).

💧 Lesson 5: Evaporation & Distillation

Key Concepts to Know:

- What happens during evaporation
- Distillation: how it separates and keeps both solute and solvent
- Why distillation is used in places like the space station
- Practical uses of both methods

Questions to Check Understanding:

1. What is evaporation?
2. What happens to the solvent during evaporation?
3. Why does distillation recover both parts?
4. What are some real-world uses of distillation?
5. Which method would astronauts prefer and why?

Revision Task:
🔁 Draw a flowchart showing the steps of **distillation** and **evaporation** and label which part is kept or lost.

🖍️ Lesson 6: Chromatography

Key Concepts to Know:

- What chromatography is and what it separates
- How the process works (paper = stationary phase, solvent = mobile phase)

- Why some colors move farther than others
- Real-world applications: forensics, food dyes

Questions to Check Understanding:

1. What does chromatography separate?
2. What moves farther in chromatography and why?
3. What are the two “phases” in paper chromatography?
4. Give two real-life uses for chromatography.
5. What is a chromatogram?

Revision Task:
🎨 Do a mini chromatography demo at home (if possible), or sketch how different dyes travel on paper.

📝 Final Practice

Short Answer Practice Questions:

1. Explain how tissues, organs, and systems are related.
2. What makes plant cells different from animal cells?
3. What separation method is best for saltwater?
4. Why is chromatography useful in solving crimes?
5. Describe one situation where distillation is more useful than evaporation.

Challenge:
Write a paragraph explaining how the parts of a cell support the life processes in MRS GREN.

✅ Revision Checklist:

Revision Task	Done?
Reviewed Canvas slides for Lessons 1–6	☐
Answered all "Check Understanding" questions	☐
Completed all diagrams and summary tables	☐
Practiced writing short answers	☐
Asked for help on anything confusing	☐

✨ You’re going to do great! Use this booklet to structure your study time and review what you know. You’ve learned so much already — just bring it all together now!